StockSense — Complete Product & Technical Documentation

Live Document — Updated automatically as the product evolves.

Last updated: 2026-06-22 (Session 7)

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1. Product Overview

StockSense is an AI-powered stock prediction and portfolio intelligence platform built for Indian and US equity markets. It combines institutional-grade quantitative signals with a consumer-friendly interface to deliver actionable BUY / HOLD / SELL signals with full explainability.

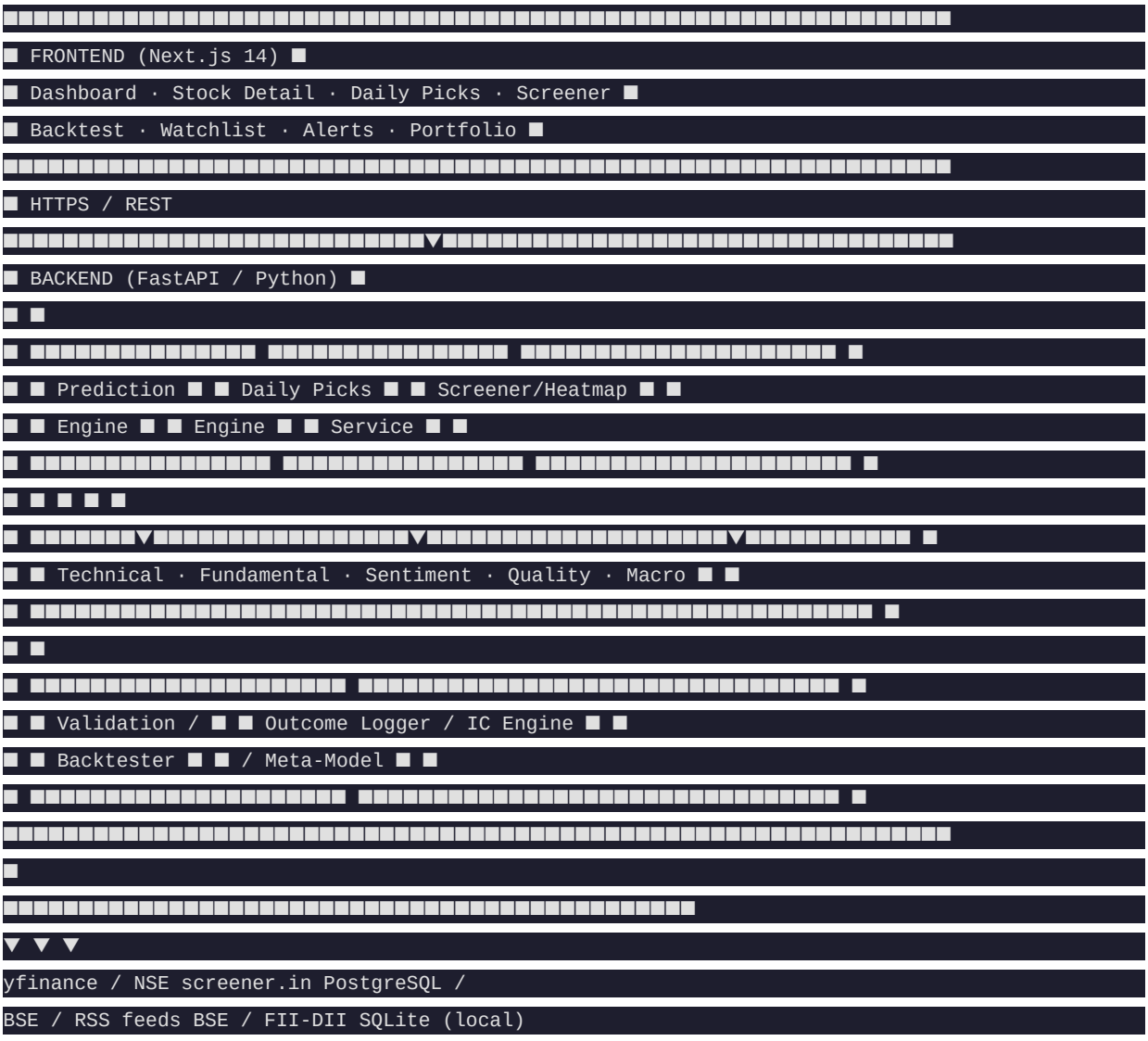
What StockSense Does

- Generates **BUY / HOLD / SELL signals** for Nifty 100, US large-cap, and top cryptocurrencies
- Delivers **Daily Picks** every morning at 9 AM IST — top 5 BUY ideas per horizon (short / medium / long)
- Shows **why** every signal was generated — factor breakdown, confidence scores, reasoning bullets
- Provides **trade levels** — entry zone, stop-loss, and target price with R:R ratio
- Runs a **learning engine** — tracks prediction outcomes and retrains factor weights weekly
- Supports **screener**, **backtest**, **watchlist**, and **alerts**

Supported Markets

Market	Universe	Horizons
India (NSE)	Nifty 100	Short (1–5 days), Medium (2–4 weeks), Long (3–6 months)
US	S&P 500 large-caps	Short, Medium, Long
Crypto	BTC, ETH, BNB, SOL, XRP, DOGE, ADA, AVAX, LINK, DOT	Short, Medium, Long

2. Architecture Overview



Tech Stack

Layer	Technology
Frontend	Next.js 14, React, TypeScript, TailwindCSS, Recharts
Backend	Python 3.11, FastAPI, Uvicorn
Auth	Supabase (JWT)
Database	PostgreSQL (production), SQLite (local)

| Cache | In-memory (TTL-based), disk (picks_cache.json) |

| Hosting | Render.com (backend), Vercel (frontend) |

| Automation | GitHub Actions (cron jobs) |

3. Data Sources

| Source | Data Provided | Frequency | Used For |

|--------|--------------|-----------|----------|

| **yfinance** | Price, OHLCV, P/E, ROE, FCF, beta, analyst targets | Real-time | All markets |

| **screener.in** | 10-year financials, ROCE, CAGR, promoter %, pledge % | Daily | India fundamentals |

| **BSE API** | Fundamentals for renamed / merged stocks | Daily | India fallback |

| **NSE FII/DII API** | Daily institutional flows (■ Cr) | Daily | India quality signal |

| **NSE Pledge API** | Promoter pledge % (quarterly disclosure) | Quarterly | India risk signal |

| **Yahoo Finance RSS** | News headlines per symbol | Real-time | Sentiment |

| **Google News RSS** | `{symbol} stock India` search results | Real-time | Sentiment fallback |

| **Economic Times RSS** | India economy & market news | Real-time | India sentiment |

| **MoneyControl RSS** | Stock & sector news | Real-time | India sentiment |

| **yfinance macro** | S&P 500, VIX, Crude, Gold, USD/INR, Nifty IT/Bank | 15-min cache | Global macro |

Data Fallback Chain (India)

```
yfinance → screener.in (if <5 key fields) → BSE API (if still sparse)
```

4. Core Prediction Engine

File: `backend/services/prediction_engine.py`

Signal Generation Formula

```
Composite Score = (Tech × W_tech) + (Fund × W_fund) + (Sent × W_sent)
+ Global Macro Adjustment
+ Analyst Consensus Adjustment
+ 52-Week Position Adjustment
+ Quality Factor Adjustment
+ Rounding Adjustment
- Risk Penalty
```

All component scores are on a **0–100 scale** (50 = neutral). The composite is also 0–100.

Signal Thresholds

| Composite Score | Signal | Score Band | Confidence Calculation |

|----------------|--------|------------|----------------------|

| ≥ 90 | **BUY** | Exceptional Opportunity | `(score − 60) / 40 × 100%` |

| ≥ 75 | **BUY** | Strong Buy Candidate | `(score − 60) / 40 × 100%` |

| ≥ 60 | **BUY** | Good Watchlist Stock | `(score − 60) / 40 × 100%` |

| 45 – 59 | **HOLD** | Neutral — Monitor | `50 − abs(score − 52) × 2` |

| < 45 | **SELL** | Avoid | `(45 − score) / 45 × 100%` |

Dynamic Weights (Horizon × Volatility × Regime)

Base weights by horizon:

Horizon	Technical	Fundamental	Sentiment
Short (1–5 days)	70%	15%	15%
Medium (2–4 weeks)	40%	45%	15%
Long (3–6 months)	15%	75%	10%

Volatility modulation (applied on top of base weights):

Volatility Level	Annualised Vol	Technical	Fundamental
High	> 35%	Reduced to 10%	Boosted
Normal	15–35%	Base	Base
Low	< 15%	Boosted to 75%	Reduced

Regime modulation:

- **BULL regime**: Boost technical, reduce fundamental
- **BEAR regime**: Boost fundamental, reduce technical
- **SIDEWAYS**: No modulation (enables mean-reversion trades)

Prediction Caching

- **TTL:** 15 minutes per `(symbol:market:horizon)` key
- **Max size:** 300 entries (Render 512 MB free-tier limit)
- **Eviction:** LRU — oldest entry dropped when full
- **Async pattern:** First request returns HTTP 202 (computing); client polls every 5s; result cached on completion

5. Technical Analysis Module

File: `backend/services/technical_indicators.py`

Indicators Computed

Category	Indicator	Parameters
Momentum	RSI	14-period
	Stochastic RSI	14-period
	Williams %R	14-period
	CCI	20-period
	Stochastic Oscillator	K=14, D=3
Trend	MACD	12/26/9 EMA
	EMA	20, 50, 200 periods
	ADX + DI+/DI−	14-period
Volatility	Bollinger Bands	20-period, 2σ
	ATR	14-period

| **Volume** | OBV | Rolling |

| | Volume SMA | 20-period |

| | VWAP | 20-day rolling |

Scoring Rules

RSI:

- < 30 (oversold): **+15 pts**
- 30–45 (recovering): **+7 pts**
- 60–70 (elevated): –**7 pts**
- > 70 (overbought): –**15 pts**

MACD:

- Above signal line: **+12 pts**
- Below signal line: –**12 pts**

EMA Trend:

- Price > EMA200 (bull market structure): **+10 pts**
- EMA20 > EMA50 (golden cross zone): **+8 pts**
- EMA20 < EMA50 (death cross zone): –**8 pts**

ADX (Trend Strength):

- ADX > 25 AND +DI > –DI (strong uptrend): **+10 pts**
- ADX > 25 AND –DI > +DI (strong downtrend): –**10 pts**
- ADX ≤ 25 (sideways): **0 pts**

Bollinger Bands (%B):

- Near lower band (< 0.1): **+8 pts** (oversold)
- Near upper band (> 0.9): –**8 pts** (overbought)

Volume Confirmation:

- Up-day volume > down-day volume: **+8 pts**
- Down-day volume > up-day volume: –**8 pts**

Candlestick Patterns:

| Pattern | Points | Signal |

|---------|--------|--------|

| Hammer | +5 | BUY |

| Morning Star | +5 | BUY |

| Bullish Engulfing | +5 | BUY |

| Shooting Star | –5 | SELL |

| Evening Star | –5 | SELL |

| Bearish Engulfing | –5 | SELL |

Technical Signal Output

- Score ≥ 58: **BUY**
- Score ≤ 42: **SELL**
- 42–58: **HOLD**

6. Fundamental Scoring Module

File: `backend/services/prediction_engine.py` → `_fundamental_score()`

Quality Gate (Hard Rejection)

A stock is flagged **REJECTED** before scoring if **any** of these apply:

- ROE < –10% (severely destroying shareholder value)
- Profit Margin < –15% (deeply loss-making)
- Non-positive Operating Cash Flow on medium/long horizons (core business not generating cash)
- D/E ratio > 500% (extreme leverage, non-financial sector only)

> Previously ROE AND margin both had to be negative simultaneously — this was too lenient and has been corrected.

Scoring Architecture (Per-Category Budgets)

The fundamental score uses **six independent capped buckets**, each scored separately, then summed with a base of 50. This prevents any single dimension from dominating and ensures the final score has meaningful discrimination across the full 0–100 range.

Bucket	Cap	What It Measures
Valuation	±15	P/E, P/B ratios vs market-calibrated thresholds
Profitability	±15	ROE, ROCE, profit margins
Growth	±15	Revenue + earnings CAGR (counted once via longest available window)
Balance Sheet	±10	D/E, OCF quality, Altman Z-Score, Sloan accruals
Governance	±10	Promoter holding, FII/DII flows, promoter pledge
Banking	±10	Net NPA, NIM (fires only for banks/NBFCs)

Total possible range: 50 ± 65 → clamped to [0, 100]

Valuation bucket (cap ±15)

Metric	Threshold (India / US)	Points
P/E	< 18 IN / < 15 US (cheap)	+8
P/E	< 30 IN / < 25 US (fair)	+3
P/E	> 55 IN / > 50 US (expensive)	–8
P/B	< 2.5 IN / < 2.0 US	+4
P/B	> 8.0 IN / > 6.0 US	–4

Profitability bucket (cap ±15)

Metric	Threshold	Points
ROE	> 20%	+7
ROE	10–20%	+3
ROE	< 0%	–7
Profit Margin	> 20%	+5
Profit Margin	< 0%	–5
ROCE	> 20%	+6
ROCE	12–20%	+2
ROCE	< 6%	–4

Growth bucket (cap ±15) — revenue and earnings each counted once

Revenue uses 3Y CAGR (screener.in) if available, else TTM YoY (yfinance):

Metric	Threshold	Points
3Y Revenue CAGR	> 15%	+7
3Y Revenue CAGR	8–15%	+3
3Y Revenue CAGR	< 0%	–5
TTM Revenue Growth (fallback)	> 20%	+7
TTM Revenue Growth	5–20%	+3
TTM Revenue Growth	< –5%	–5

Earnings uses longest available: 5Y CAGR (long horizon) → 3Y CAGR → TTM EPS growth:

Metric	Threshold	Points
5Y Profit CAGR (long only)	> 18%	+6
3Y Profit CAGR	> 20%	+6
3Y Profit CAGR	> 10%	+3
3Y Profit CAGR	< –10%	–5
TTM EPS Growth (fallback)	> 20%	+5
TTM EPS Growth	< –10%	–5
Quarterly PAT trend	Accelerating	+3
Quarterly PAT trend	Decelerating	–3

Balance Sheet bucket (cap ±10)

Metric	Threshold	Points
D/E	> 300%	–7
D/E	150–300%	–3
D/E	< 50%	+3
Operating CF (screener)	Negative	–5
Operating CF 3Y growth	> 30%	+4
Operating CF 3Y growth	Positive	+2
Altman Z-Score	Safe zone (medium/long)	+3
Altman Z-Score	Grey zone	–4
Altman Z-Score	Distress zone	–8
Sloan Accruals ratio	< –5% (cash-backed)	+3
Sloan Accruals ratio	> 10% (manipulation risk)	–5

Governance bucket (cap ±10) — India only

Metric	Threshold	Points
FII + DII combined	> 50%	+4
FII + DII combined	25–50%	+2
DII quarterly trend	Up > 3% (MF accumulation)	+3

| DII quarterly trend | Down > 3% | –3 |

| FII quarterly trend | Up > 3% | +2 |

| FII quarterly trend | Down > 3% | –2 |

| Promoter holding | > 55% | +2 |

| Promoter holding | < 25% | –2 |

| Promoter trend | Up > 2% (insider buying) | +3 |

| Promoter trend | Down > 3% (insider selling) | –4 |

| Promoter pledge | > 50% | –8 |

| Promoter pledge | 25–50% | –5 |

| Promoter pledge | 10–25% | –2 |

| Promoter pledge | 0% | +2 |

Banking bucket (cap ±10) — fires only for banks/NBFCs

| Metric | Threshold | Points |

|--------|-----------|--------|

| Net NPA | > 3% | –7 |

| Net NPA | 1.5–3% | –3 |

| Net NPA | < 0.5% | +4 |

| NIM | > 4% | +4 |

| NIM | < 2% | –3 |

7. Sentiment Analysis Module

File: `backend/services/news_sentiment.py`

Two-Layer Scoring

Layer 1 — Financial Lexicon (60% weight):

- ~80 domain-specific phrases with pre-calibrated scores
- Examples: `"beat expectations"` → +0.75, `"profit warning"` → –0.80, `"upgrade"` → +0.65
- Designed to override generic sentiment on financial language

Layer 2 — VADER Sentiment (40% weight):

- Title sentiment (70%) + Description sentiment (30%)
- Standard NLP library tuned for social media / news

Final blend: `score = 0.60 × lexicon + 0.40 × VADER`

News Sources (per prediction)

| Source | Articles | Priority |

|--------|----------|----------|

| Yahoo Finance RSS (`{symbol}`, `{symbol}.NS`) | 10 | Primary |

| Google News RSS (`{symbol} stock India`) | 10 | Secondary |

| Economic Times RSS | 10 | India supplement |

| MoneyControl RSS | 10 | India supplement |

Cache TTL: 10 minutes per symbol

Sentiment Classification

| Score | Label |

|-------|-------|

| ≥ +0.05 | BULLISH |

| –0.05 to +0.05 | NEUTRAL |

| ≤ –0.05 | BEARISH |

Score conversion: `sentiment_score = 50 + (bullish − bearish) / (bullish + bearish) × 50`

> Neutral articles are excluded from the denominator. Previously neutrals diluted the score — 5 bullish + 5 neutral incorrectly scored the same as 5 bullish + 5 bearish. Now only labelled articles (bullish + bearish) count.

When no news available: Returns neutral (50), redistributes weight to technical + fundamental, sets `data_available = False` flag.

8. Global Macro Context Module

File: `backend/services/global_context.py`

Macro Indicators Tracked

| Indicator | Ticker | What It Signals |

|-----------|--------|-----------------|

| S&P 500 | ^GSPC | US demand → IT/pharma export tailwind |

| NASDAQ | ^IXIC | Tech hiring, demand for Indian IT services |

| VIX | ^VIX | FII flows: high VIX → FII outflows from India |

| USD/INR | INR=X | Weak INR → pharma/IT revenue boost; import cost rise |

| Crude Brent | BZ=F | Oil import cost, OMC margins |

| Gold | GC=F | Jewelry demand (Titan, Muthoot) |

| Nifty IT Index | ^CNXIT | Sector momentum for IT stocks |

| Nifty Bank Index | ^NSEBANK | Banking sector rotation signal |

Cache TTL: 15 minutes (fetched in parallel)

Stock-Specific Macro Sensitivity Map

Over 100 Nifty stocks mapped to their macro sensitivities:

| Stock Category | Tailwind Factors | Headwind Factors |

|---------------|-----------------|-----------------|

| IT (TCS, INFY, WIPRO, HCL) | USD/INR weakness, S&P 500 up, NASDAQ up | INR strengthening |

| Pharma (SUNPHARMA, DRREDDY) | USD/INR weakness, S&P 500 up | — |

| OMCs (BPCL, IOC) | Crude down | Crude up (margin squeeze) |

| Oil producers (ONGC) | Crude up | Crude down |

| Paints (ASIANPAINT, PIDILITIND) | — | Crude up (TiO2 input cost) |

| Banks (HDFCBANK, ICICIBANK) | — | VIX up (FII sensitivity) |

| Jewelry (TITAN) | Gold stable | Gold up (input cost) |

Stock Adjustment Calculation

```
stock_adj = Σ sensitivity_i × (factor_i − benchmark_i)
Tailwind factors: +2 to +4 points
Headwind factors: −2 to −4 points
```

9. Quality Factors Module

File: `backend/services/quality_factors.py`

> Applied for **India (Nifty 100) only**, on medium and long horizons.

10 Dimensions of Quality

1. Earnings Revisions (Weight: 12–14%)

- EPS surprise trend (beat vs miss last 4 quarters): ±16 pts
- Analyst upgrade/downgrade momentum: ±8 pts
- Forward PE compression vs trailing PE: ±8 pts

2. Institutional Ownership (Weight: 5–6%)

- Holdings > 50%: +14 pts
- Holdings 30–50%: +8 pts
- Holdings 15–30%: +3 pts
- Holdings < 5%: –5 pts
- Institution count > 300: +6 pts; < 20: –4 pts

3. Institutional Flow Proxy / MF Trend (Weight: 5–13%)

Blends price-volume signals (OBV trend, MFI, accumulation pattern) with real NSE FII/DII data:

- 60% price-volume proxy + 40% real FII/DII flows (when available)

4. Relative Strength (Weight: 7–15%)

Stock return vs NIFTY 50 over 1M, 3M, 6M:

- Outperform > 10%: +12 pts
- Outperform 4–10%: +6 pts
- Underperform > 10%: –12 pts
- Underperform 4–10%: –6 pts

5. Sector Strength (Weight: 7–15%)

Sector index momentum vs NIFTY 50:

- Sector outperform > 5%: +16 pts
- Sector outperform 2–5%: +8 pts
- Sector underperform > 5%: –14 pts
- Sector underperform 2–5%: –7 pts

6. Valuation Quality (Weight: 8–17%)

Multi-dimensional valuation scoring:

- PEG < 0.75: +16 pts; > 2.5: –12 pts
- EV/EBITDA vs sector: ±10 pts at major discount/premium
- Sector-relative PE: ±12 pts at 30% discount
- P/B for banks < 1.0: +12 pts; > 4.0: –8 pts
- Analyst target upside > 30%: +10 pts (margin of safety)

7. Risk Management (Weight: 10%)

- Max Drawdown 12M < –10%: +14 pts; < –40%: –14 pts
- Volatility Percentile bottom 25%: +10 pts; top 20%: –10 pts
- Sharpe Ratio > 1.5: +12 pts; < 0: –10 pts

- Downside Deviation < 10%: +6 pts; > 30%: –6 pts

8. Corporate Actions (Weight: 3–10%)

- Dividend payer 5Y+: +8 pts
- Growing dividends: +6 pts
- Payout ratio 0–40%: +4 pts; > 80%: –6 pts
- Active buyback: +8 pts
- Share dilution: –5 pts

9. Liquidity (Weight: 4–8%)

- Market cap ■2T+: +10 pts; < ■10B: –8 pts
- Avg daily volume > 5M: +8 pts; < 100K: –8 pts
- Beta 0.5–1.2: +4 pts; > 2.0: –5 pts

10. Quality Metrics (Weight: 4–12%)

Piotroski F-Score (9-point scale):

Point	Criterion
P1	ROA > 0
P2	Operating Cash Flow > 0
P3	ROA improving YoY
P4	Accruals: cash earnings > accrual earnings
P5	Leverage decreasing
P6	Current ratio improving
P7	No share dilution
P8	Gross margin expanding
P9	Asset turnover improving

- Score ≥ 8: +20 pts
- Score 6–7: +10 pts
- Score 4–5: 0 pts
- Score < 4: –12 pts

ROIC (Return on Invested Capital):

- > 20%: +14 pts
- 12–20%: +8 pts
- 6–12%: +2 pts
- < 6%: –8 pts

Horizon-Based Weighting

Dimension	Short	Medium	Long
Earnings Revisions	13%	14%	12%
Institutional Ownership	5%	6%	6%
MF/FII Flow	6%	8%	10%
Relative Strength	15%	11%	7%
Sector Strength	15%	11%	7%
Valuation Quality	8%	13%	17%
Risk Management	10%	10%	10%

| Corporate Actions | 3% | 6% | 10% |

| Liquidity | 8% | 6% | 4% |

| Quality Metrics | 4% | 6% | 12% |

| Flow Proxy | 13% | 9% | 5% |

10. Risk Penalty Framework

Applied **after** all signal scoring. Never adds points — only subtracts (risk override).

| Risk Factor | Condition | Penalty |

|---|---|---|

| High leverage | D/E > 300% | –8 pts |

| Elevated leverage | D/E 200–300% | –4 pts |

| High beta | Beta > 2.0 | –6 pts |

| Elevated beta | Beta 1.6–2.0 | –3 pts |

| Negative FCF | FCF < 0 | –5 pts |

| Negative ROE | ROE < 0 | –5 pts |

| Poor risk profile | Risk score < 35 | –5 pts |

| Earnings volatility | CV > 0.5 | –4 pts |

Maximum penalty capped at –30 pts (prevents extreme scores on genuinely bad risk/reward stocks).

11. Confidence Engine

Answers: **"How much should you trust this signal?"**

Five Components

| Component | What It Measures | Weight (Full) | Weight (Bootstrap) |

|---|---|---|---|

| Data Completeness | % of key fundamental fields present | 25% | 31.25% |

| Factor Agreement | % of factors agreeing with signal direction | 25% | 31.25% |

| Earnings Stability | Quality earnings_revision sub-score | 15% | 18.75% |

| Regime Certainty | BULL/BEAR vs SIDEWAYS trend strength | 15% | 18.75% |

| Historical Factor Reliability | Live IC values (needs 60+ outcomes) | 20% | 0% → fallback 50 |

Sector-Aware Field Sets

Indian Non-Financial: PE, ROE, revenue growth, D/E, margin, beta (yfinance) + 3Y CAGR, ROCE, FII, promoter (screener) = 10 fields

Indian Financial (banks/NBFCs): PE, ROE, revenue growth, profit margin, EPS growth, beta (yfinance) + 3Y CAGR, FII, promoter (screener) = 10 fields

US Stocks: PE, ROE, revenue growth, D/E, margin, EPS growth, FCF, beta, ROCE = 13 fields

Confidence Bands

| Score | Band |

|-------|------|

| ≥ 80 | **HIGH** |

| 60–79 | **MEDIUM** |

| < 60 | **LOW** |

Percentile Context

| Score | Label |

|-------|-------|

| ≥ 80 | Top 10% of all Nifty predictions |

| ≥ 72 | Top 20% |

| ≥ 65 | Top 35% |

| ≥ 58 | Top 50% |

| ≥ 50 | Below average |

| < 50 | Low range |

12. Target Price & Trade Levels

File: `backend/services/prediction_engine.py` → `_target_price()`, `_trade_levels()`

Target Price

| Horizon | Method |

|---------|--------|

| **Short** | ATR × 2.5 × confidence factor — moves 1–5 day magnitude |

| **Medium** | Analyst target (70%) + price projection (30%); BUY floor: `max(blend, price×1.05)` |

| **Long** | `analyst_target × (1+EPS_growth)²`; BUY floor: `max(target, price×1.15)` |

Trade Levels

```
Take Profit = Model Target Price (consistent with signal)
Stop Loss = ATR-based, adjusted to maintain minimum R:R of 1.5×
Entry Zone = [price − 0.3×ATR, price + 0.1×ATR] for BUY
[price − 0.1×ATR, price + 0.3×ATR] for SELL
R:R Ratio = (target − price) / (price − stop_loss)
```

ATR multipliers by horizon:

| Horizon | Base SL | Floor SL | Max SL |

|---------|---------|----------|--------|

| Short | 1.5× ATR | 1× ATR | 25% of price |

| Medium | 3× ATR | 2× ATR | 25% of price |

| Long | 5× ATR | 3× ATR | 25% of price |

13. Daily Picks Engine

File: `backend/services/daily_picks.py`

Execution Schedule

- Triggered: **Every weekday at 9:00 AM IST** (via GitHub Actions → POST `/api/picks/generate`)
- Generation time: ~10 minutes
- Results cached to disk: `backend/picks_cache.json`
- API response: Instant (reads from cache)

9-Phase Pipeline

Phase 0 — Outcome Resolution

- Compare previous predictions against actual forward returns (1-day / 63-day / 252-day)
- Update outcome logger database with direction hits (correct/incorrect)
- Feed data into IC engine for retraining

Phase 1 — Universe Screening

- Run full prediction engine on all Nifty 100 stocks
- Parallelised: 2 workers via `ThreadPoolExecutor`
- Returns raw factor scores for all stocks (enables cross-sectional z-scoring)

Phase 2 — Regime Detection

- KMeans clustering (4 clusters) on global macro features: VIX, S&P 500 return, crude, gold, USD/INR
- Classifies market into: **BULL_CALM**, **BULL_VOLATILE**, **BEAR_CALM**, **BEAR_PANIC**
- Returns regime label + weight multipliers for IC adjustment

Phase 3 — IC Weight Computation

- If < 60 outcome pairs: use academic prior weights

- Short: tech=0.055, fund=0.018, sentiment=0.042, quality=0.032

- If ≥ 60 pairs: Bayesian shrinkage blend of live IC + prior:

- `weight = live_weight × live_ic + (1 − live_weight) × prior`

- Apply regime multipliers: BULL boosts tech/sentiment; BEAR boosts fundamental/quality

Phase 4 — Z-Score Normalisation & Alpha

- Cross-sectional z-scoring: `z_i = (score_i − mean) / std`
- Combined alpha: `Σ ic_weight_k × z_k`
- Meta-model alpha (requires 180+ outcomes across 60 stocks × 3 horizons):

- Inputs: tech_z, fund_z, sentiment_z, quality_z, combined_alpha, regime_id

- Output: predicted return %

Phase 5 — Pick Selection

- Rank by alpha score (meta_alpha if available, else combined_alpha)
- Select top 5 **BUY** signals per horizon (composite score ≥ 60)
- Minimum 1 pick per horizon
- Empty picks from a prior run (0 BUY signals) are NOT treated as "complete" — startup catch-up will retry on next deploy

Phase 6 — Portfolio Optimisation

- Fetch 6-month daily returns for selected picks
- Covariance estimation: Ledoit-Wolf shrinkage at 25%
- Optimise: `max (alpha × w − λ × w^T Σ w)` via SLSQP
- Constraints: `Σw = 1.0`, `0 ≤ w_i ≤ 0.40` (max 40% per position)
- Risk aversion `λ`: doubled in BEAR_PANIC regime
- Fallback (if scipy unavailable): iterative alpha-proportional weights that correctly enforce the 40% cap

Phase 7 — Logging

- Log to PostgreSQL (if `USE_POSTGRES=1`) or SQLite
- Stored fields: factor z-scores, combined_alpha, meta_alpha, signal, price, horizon, regime

Phase 8 — Weight Adaptation (background)

- Retrain IC engine with new outcomes
- Update meta-model if sufficient data
- Recalibrate regime clustering
- Runs in daemon thread (non-blocking)

14. Backtesting & Validation Engine

File: `backend/services/validation_engine.py`, `backend/services/backtester.py`

Walk-Forward Methodology

For each business day `t` in Nifty 100 history:

1. Fetch price data available **only before** time `t`
2. Compute prediction at `t` using that data
3. Measure actual forward return at `t + h` (h = 7 / 63 / 252 days)
4. Compare predicted signal vs actual direction

> **Known limitation (next session):** Technical indicators (EMA, MACD, OBV) are currently computed on the full historical DataFrame before slicing per date — a form of look-ahead bias in the backtester. This inflates reported validation hit rates. A full fix (recompute indicators per window) is planned for Session 5.

Metrics Computed

| Metric | Definition |

|--------|-----------|

| **Hit Rate** | % of BUY/SELL calls that were directionally correct |

| **Avg Return on BUY** | Mean forward return when signal was BUY |

| **Sharpe Ratio** | `(avg_return – risk_free) / std_return` on BUY calls |

| **vs Benchmark** | Alpha over NIFTY 50 buy-and-hold |

| **Score Calibration** | Hit rate by score bucket (60–70, 70–80, 80–100) |

Execution

- **Weekly:** Sunday 7:30 AM IST (via GitHub Actions)
- **Duration:** ~40 minutes (medium + long horizons)
- **Storage:** PostgreSQL `val_runs` + `val_signals` tables
- **Retention:** 365 days

Forward Return Windows

| Horizon | Forward Return Used | Outcome Resolution Wait |

|---------|---------------------|-------------------------|

| Short | return_5d (5 trading days) | 3 calendar days |

| Medium | return_20d (20 trading days ≈ 1 month) | 30 calendar days |

| Long | return_60d (60 trading days ≈ 3 months) | 90 calendar days |

> Partial returns are never logged — the outcome logger returns None if fewer than the required trading days have elapsed. This prevents truncated returns from contaminating IC training data.

Validation BUY Threshold

Validation uses the same threshold as the live prediction engine: **composite ≥ 60 = BUY** for all horizons. Previously the thresholds were mismatched (validation used 65, live used 60), making validation metrics unmeasurable against the actual model.

Indicative Score Calibration (Nifty 100, Medium-Term)

Score Bucket	Hit Rate	Beat Benchmark	Avg Alpha
80–100 (Exceptional/Strong BUY)	~68%	~62%	+4.2%
75–79 (Strong BUY)	~62%	~55%	+2.8%
60–74 (Good Watchlist BUY)	~58%	~50%	+1.5%
45–59 (HOLD)	~52%	~48%	+0.3%

> Note: These figures reflect the post-Session 4 thresholds. Historical validation data accumulated under the old 70-threshold is being re-calibrated.

15. Crypto Prediction Module

File: `backend/services/crypto_engine.py`

Supported Assets

BTC, ETH, BNB, SOL, XRP, DOGE, ADA, AVAX, LINK, DOT

Signals Used (No Fundamentals)

Signal	Weight
Technical indicators (full suite)	Primary
Fear & Greed proxy (30-day vol / 90-day vol)	Secondary
On-chain proxy (price-volume accumulation)	Secondary
News sentiment (Google News + CoinTelegraph)	Supporting
Macro sensitivity (BTC/ETH correlated to S&P, VIX)	Adjustment

> Quality factors and dynamic fundamental weights are not applied — crypto lacks P/E, ROE, cash flow data. Momentum and sentiment dominate.

16. Screener & Universe Management

File: `backend/services/screener_service.py`, `screener_data.py`

Screener Filters

Filter	Type	Description
Market	US / IN	Filter by market
Signal	BUY / HOLD / SELL	Filter by signal
Min Market Cap	■ Cr	Size filter
Max P/E	Ratio	Valuation filter

| Min ROE | % | Profitability filter |

| Sector | IT / Banking / Pharma / etc. | Sector filter |

Heatmap

- Groups stocks by sector; sorted by sector avg change% (best sectors on top)
- **India: 25 sectors** — Banking, IT, Auto, Pharma, Energy, FMCG, Finance, Healthcare, Insurance, Chemicals, Cement, Metal & Mining, Defence, Realty, Telecom, Consumer Disc, Hotels & Travel, Food & Beverage, Media & Entmt, Textiles, Agro & Chemicals, Logistics, Paints, Infra, Capital Goods, Power, EV & New Energy
- **US: 29 sectors** — Mega Cap Tech, Semiconductors, Cloud & SaaS, Cybersecurity, Fintech, Finance, Insurance, Healthcare, Biotech, Med Devices, Energy, Clean Energy, EV, Consumer Disc, Consumer Stap, E-commerce, Social Media, Streaming & Media, Gaming, Aerospace & Defence, Industrials, Airlines, Cruise & Hotels, Restaurants, Retail, Telecom, Utilities, Realty, Materials, Crypto & Blockchain
- Up to **15 stocks per sector** (MAX_STOCKS = 15)
- Primary data source for India sectors: **NSE sector index APIs** (`SECTOR_TO_NSE_INDEX` mapping); Yahoo Finance used as fallback for symbols not in NSE indices and as primary source for all US sectors
- Grey tiles = symbol has no live data; all 353 India symbols audited and bad tickers corrected (Session 5)
- Colour-coded by performance (green/red intensity); loading status badge (Fetching / Refreshing / Live)

Top Movers

- Fetched in real-time from yfinance
- Ranked by absolute % change
- Separate for US, India, Crypto

17. Paper Trading Module

File: `backend/api/routers/paper_trading.py`, `frontend/src/app/paper-trading/page.tsx`

A simulated trading environment where users can test stock calls without real money. All trades persist in PostgreSQL and survive Railway restarts.

Database Schema

```
paper_portfolio (user_id TEXT PK, session_id TEXT, cash NUMERIC, updated_at TIMESTAMPTZ)
paper_trades (id SERIAL PK, user_id TEXT, session_id TEXT, symbol TEXT, market TEXT,
quantity INT, entry_price NUMERIC, exit_price NUMERIC,
stop_loss NUMERIC, target_price NUMERIC, status TEXT,
signal TEXT, horizon TEXT, opened_at TIMESTAMPTZ, closed_at TIMESTAMPTZ)
```

User Model

- Trades are scoped to **Supabase** `user_id` (stable UUID from `useAuth().user.id`)
- No more session_id / localStorage dependency — trades persist across all browsers and devices
- Starting virtual cash: **₹1,00,000 / $10,000** (depending on market)
- All positions are long only (no shorting)

Trade Lifecycle

```
Open Trade → entry_price captured at trade time, keyed to user_id
Live Price → fetched real-time via yfinance / NSE
Unrealised P&L = (live_price − entry_price) × quantity
Close Trade → exit_price set, status = 'CLOSED'
```

```
Realised P&L = (exit_price − entry_price) × quantity
```

Stop Loss & Target Price

- Optionally set per trade via inline edit (✎ icon)
- ATR-based defaults pre-filled when placing a trade from the stock detail page
- Both values always shown per position with % from entry
- UI highlights rows where price is within 2% of stop (yellow) or target (green)
- The Buy button is **not blocked** by AI prediction loading — trade is placeable immediately; stop loss/target suggestions update in the background

API Endpoints

| Endpoint | Method | Description |

|---|---|---|

| `/api/paper-trading/portfolio` | GET | Portfolio summary (pass `user_id` as query param) |

| `/api/paper-trading/buy` | POST | Open a new position (body: `user_id`, symbol, market, qty, price, …) |

| `/api/paper-trading/sell/{trade_id}` | POST | Close a position (body: `user_id`, price) |

| `/api/paper-trading/trade/{trade_id}` | PATCH | Edit stop_loss / target_price (body: `user_id`, …) |

| `/api/paper-trading/reset` | POST | Reset portfolio to starting cash (query: `user_id`) |

18. Alerts System

File: `backend/api/routers/alerts.py`, `frontend/src/app/alerts/page.tsx`

Price-level alerts that trigger when a stock crosses a target price. All alerts persist in PostgreSQL.

Database Schema

```
price_alerts (id TEXT PK, user_id TEXT, symbol TEXT, market TEXT,
target_price NUMERIC, direction TEXT CHECK('above','below'),
triggered BOOL DEFAULT FALSE, created_at TIMESTAMPTZ, triggered_at TIMESTAMPTZ)
```

Alert Check Logic

- Frontend polls live quote every 60 seconds
- Compares `live_price` vs `target_price` by direction (`above` / `below`)
- On trigger: PATCH alert to `triggered = true`, stores `triggered_at`
- Triggered alerts shown with timestamp; can be reset or deleted

API Endpoints

| Endpoint | Method | Description |

|---|---|---|

| `/api/alerts/{user_id}` | GET | All alerts for user |

| `/api/alerts/{user_id}` | POST | Create new alert |

| `/api/alerts/{user_id}/{alert_id}` | PATCH | Update (reset triggered, edit price) |

| `/api/alerts/{user_id}/{alert_id}` | DELETE | Remove alert |

19. API Reference

Prediction

| Endpoint | Method | Description |

|---|---|---|

| `/api/predictions/{symbol}` | GET | Get prediction (async, 202 while computing) |

| `/api/predictions/debug/state` | GET | Internal cache/thread state (debug only) |

Query params: `market` (US/IN/CRYPTO), `horizon` (short/medium/long)

Daily Picks

| Endpoint | Method | Description |

|---|---|---|

| `/api/picks/daily` | GET | Today's cached picks (instant) |

| `/api/picks/generate` | POST | Trigger pick generation (secret-protected) |

| `/api/picks/performance` | GET | Live P&L of past picks vs benchmark |

Query params for performance: `horizon` (short/medium/long), `window_days` (default 90)

Screener

| Endpoint | Method | Description |

|---|---|---|

| `/api/screener/filter` | GET | Filter universe by fundamentals |

| `/api/screener/heatmap` | GET | Sector heatmap |

| `/api/screener/top-movers` | GET | Top 10 movers by % change |

| `/api/screener/crypto-movers` | GET | Top 10 crypto movers |

Backtest & Validation

| Endpoint | Method | Description |

|---|---|---|

| `/api/backtest/run` | POST | Single-stock walk-forward backtest |

| `/api/validation/run` | POST | Full universe validation (background) |

| `/api/validation/status` | GET | Validation progress |

| `/api/validation/results` | GET | Validation metrics by horizon |

Watchlist

| Endpoint | Method | Description |

|---|---|---|

| `/api/watchlist/{user_id}` | GET | All saved stocks |

| `/api/watchlist/{user_id}` | POST | Add stock |

| `/api/watchlist/{user_id}/{symbol}` | DELETE | Remove stock |

Alerts

| Endpoint | Method | Description |

|---|---|---|

| `/api/alerts/{user_id}` | GET | All price alerts |

| `/api/alerts/{user_id}` | POST | Create alert |

| `/api/alerts/{user_id}/{alert_id}` | PATCH | Update alert (reset / edit) |

| `/api/alerts/{user_id}/{alert_id}` | DELETE | Delete alert |

Paper Trading

| Endpoint | Method | Description |

|---|---|---|

| `/api/paper-trading/portfolio` | GET | Portfolio summary (`?user_id=`) |

| `/api/paper-trading/buy` | POST | Open position |

| `/api/paper-trading/sell/{trade_id}` | POST | Close position |

| `/api/paper-trading/trade/{trade_id}` | PATCH | Edit SL / target |

| `/api/paper-trading/reset` | POST | Reset portfolio (`?user_id=`) |

Infrastructure

| Endpoint | Method | Description |

|---|---|---|

| `/health` | GET | Health check (keep-alive ping) |

20. Frontend Pages & Components

Pages

| Page | Route | Description |

|---|---|---|

| Landing | `/` (unauthenticated) | Public marketing page — features, how-it-works, CTA adapts to login state |

| Dashboard | `/dashboard` | Movers (US/IN/Crypto), market status, live index bar with loading badge |

| Stock Detail | `/stock/:symbol` | Prediction, trade levels, factor breakdown, news, chart |

| Daily Picks | `/picks` | Top BUY ideas by horizon, portfolio weights, trust layer |

| Screener | `/screener` | Filter and explore the universe |

| Backtest | `/backtest` | Single-stock historical walk-forward test |

| Heatmap | `/heatmap` | Sector colour-coded snapshot (IN / US) with loading badge |

| Watchlist | `/watchlist` | Saved stocks with live prices and change% — user-scoped |

| Alerts | `/alerts` | Price alerts with live trigger detection — user-scoped |

| Portfolio | `/portfolio` | Holdings with BUY/HOLD/SELL per position |

| Paper Trade | `/paper-trading` | Simulated trading — open/close positions, P&L — user-scoped |

| Validation | `/validation` | Hit rate, Sharpe, alpha vs benchmark |

Key Components

| Component | Description |

|----------|------------|

| `FactorAttributionWaterfall` | Score decomposition bar chart; click to drill down |

| `ConfidenceMeter` | Colour-coded progress bar (0–100) |

| `ConfidenceBreakdown` | SVG gauge + 5 confidence bars with tooltips |

| `BullBearCase` | Analyst-style bull/bear thesis bullets |

| `TradingViewWidget` | Embedded chart (visual only; not wired to engine) |

| `IndexBar` | NIFTY 50, SENSEX, VIX live strip |

| `SignalBadge` | BUY / HOLD / SELL badge with colour |

| `ScoreHistoryChart` | Composite score trend over time |

| `NewsCard` | Sentiment-tagged news card |

| `BacktestPanel` | Walk-forward results per horizon on Picks page |

| `LivePerformanceTracker` | Per-pick P&L — entry, return%, alpha vs Nifty |

| `PaperTradeModal` | Trade form (qty, horizon, pre-filled SL/target) |

Daily Picks Trust Layer

The Picks page has a collapsible **"Show Real Accuracy"** panel with three layers:

1. **Backtest results** — real walk-forward hit rate, avg return, Sharpe, alpha vs Nifty per horizon
2. **Confidence calibration table** — empirical hit rate per score band (60–65, 65–70, 70–75, 75–80, 80–85, 85–91) so users can see if higher confidence = higher win rate
3. **Live P&L tracker** — every past daily pick with entry price, current return%, and alpha vs benchmark

Pick Card UI

- Rank badge (#1–#5)
- Score band label (STRONG BUY / BUY / HOLD) with colour
- Sector tag
- Top 3 signals inline (▲ BULLISH, ▼ BEARISH, → NEUTRAL) without needing to expand
- Compact market regime bar

21. Infrastructure & Deployment

Hosting

| Layer | Provider | Plan |

|-------|----------|------|

| Backend API | Railway | Hobby ($5/month, always-on, no cold starts) |

| Frontend | Vercel | Hobby |

| Database | PostgreSQL (Railway) or SQLite (local) | — |

| Auth | Supabase | Free tier |

> **Migrated from Render to Railway** (Session 5) — Railway Hobby plan eliminates the free-tier cold-start problem (30-second spin-up delays) and provides a persistent, always-on server.

Environment Variables

| Variable | Description | Default |

|----------|-------------|---------|

| `ENVIRONMENT` | `production` / `development` | `development` |

| `DATABASE_URL` | PostgreSQL connection string | SQLite fallback |

| `USE_POSTGRES` | `1` = Postgres, `0` = SQLite | `0` |

| `PICKS_SECRET` | Secret header for `/api/picks/generate` | Required in prod |

| `PICKS_UNIVERSE_LIMIT` | Cap stock count for picks run | 25 |

| `PICKS_CANDIDATES` | Top N from Phase-0 momentum screen for deep prediction | **751** (set in Railway) |

| `SCREEN_BATCH_SIZE` | NSE bulk download batch size (memory safety) | 300 |

| `MIN_MCAP_CR` | Min market cap in ■ Cr for NSE universe | 100 |

| `SCREENER_EMAIL` | screener.in login (Indian fundamentals) | Required |

| `SCREENER_PASSWORD` | screener.in password | Required |

| `FRONTEND_URL` | Vercel frontend URL for CORS | Must be set in prod |

Backend Startup Sequence

1. Uvicorn starts FastAPI app
2. CORS middleware enabled (frontend HTTPS origin)
3. **Postgres schema initialised** (`init_db()` — creates all tables if not exist)
4. **screener.in login** — authenticated session established on boot (not lazily)
5. Prediction router with daemon-thread async pattern initialised
6. Universe refresh (async, 30s delay)
7. Keepalive loop (14-minute self-ping interval)
8. Outcome resolver (6-hour interval)
9. Warmup loop — pre-computes RELIANCE:IN:medium and AAPL:US:medium (150s delay, 90s gap)

22. Automation Workflows

Directory: `.github/workflows/`

daily_picks.yml — Daily Picks Generation

```
Schedule: 0 3 * * 1-5 (3:00 AM UTC = 8:30 AM IST, Mon–Fri)
Step 1: Poll /health up to 10× (30s gap) until Render responds 200
— prevents silent failure when server is cold-starting
Step 2: POST /api/picks/generate with x-secret header
Purpose: Ensures picks are generated even after overnight sleep
Result: Picks ready by 9:00–9:10 AM IST
```

weekly_validation.yml — Model Accuracy Validation

```
Schedule: 0 2 * * 0 (2:00 AM UTC = 7:30 AM IST, Sunday)
Steps: 1. Ping /health to wake Render
2. POST /api/validation/run?horizon=medium → wait 15 min
3. POST /api/validation/run?horizon=long → wait 25 min
4. GET /api/validation/results → print summary
Output: BUY hit rate %, alpha %, Sharpe per horizon
```

keep_alive.yml — Render Free Tier Keep-Alive

```
Schedule: */10 * * * * (every 10 minutes, 24/7)
```

```
Action: GET /health
Purpose: Prevent Render free tier from spinning down
```

23. Persistence & Data Durability

Render's free tier uses ephemeral disk — files written locally are wiped on every restart/redeploy. All user-facing and learning data is stored in PostgreSQL to survive this.

What Lives in Postgres

Data	Table	Survives Restart
Price alerts	`price_alerts`	■ Yes
Watchlist	`watchlist`	■ Yes
Paper trades & portfolio	`paper_trades`, `paper_portfolio`	■ Yes
Daily picks cache	`daily_picks_cache`	■ Yes
Validation results	`val_runs`, `val_signals`	■ Yes
Alpha engine predictions	`predictions`	■ Yes
Outcome resolution	`outcomes`	■ Yes
IC history	`factor_ic_history`	■ Yes
Regime log	`regime_log`	■ Yes
Score snapshots	`score_snapshots`	■ Yes

What Is Transient (acceptable)

Data	Storage	Why Acceptable
Trained ML models (`meta_model_*.pkl`, `regime_kmeans.pkl`)	Local file	Auto-retrains from Postgres on next run — one cycle of degraded weights, no user data lost
API response caches (quotes, heatmap, movers)	In-memory (TTL)	Market data; freshly fetched anyway

screener.in Session

- Login fires at Render boot (not lazily on first request)
- Session refreshed every 6 hours
- `SCREENER_EMAIL` + `SCREENER_PASSWORD` must be set as Render environment variables
- Login logs: `[startup] screener.in login succeeded/failed` — check Render logs after deploy

24. Factor Weights by Horizon

Short-Term (1–5 days)

Factor	Weight	Rationale
Technical	70%	Momentum and price action dominate short windows
Fundamental	15%	Slow-moving signal, limited short-term relevance
Sentiment	15%	News-driven price moves matter in 1–5 days

| Quality | Not applied | Computation overhead; not predictive short-term |

Medium-Term (2–4 weeks)

| Factor | Weight | Rationale |

|--------|--------|-----------|

| Technical | 40% | Trend still matters but less dominant |

| Fundamental | 45% | Growth and valuation start to drive returns |

| Sentiment | 15% | News flow still relevant over multi-week horizon |

| Quality | Included in composite | Institutional signals begin to matter |

Long-Term (3–6 months)

| Factor | Weight | Rationale |

|--------|--------|-----------|

| Technical | 15% | Mean reversion reduces technical edge |

| Fundamental | 75% | Business quality and valuation dominate |

| Sentiment | 10% | Structural, not ephemeral, news matters |

| Quality | Included, weighted heavily | Piotroski, ROIC, cashflow most predictive |

25. Key Design Principles

1. **No Look-Ahead Bias** — Backtester uses only data available at the prediction date. Forward returns computed strictly after prediction timestamp.
2. **Full Explainability** — Every signal has reasoning bullets, factor breakdown, and confidence components. Nothing is a black box.
3. **Honest Risk-Adjusted Returns** — Target prices are not inflated. Trade levels (entry, stop, target) are mutually consistent with the signal.
4. **Sector-Aware Scoring** — Different valuation thresholds for banks (ROE > 15%, P/B < 1.5), NBFCs, and IT companies vs industrials.
5. **Institutional-Grade Signals** — Piotroski F-Score, Sharpe Ratio, ROIC, IC engine, Ledoit-Wolf covariance optimisation — same tools used by quant funds.
6. **Data Resilience** — Three-layer fallback chain: yfinance → screener.in → BSE API. If news is unavailable, weights redistribute gracefully.
7. **Memory-Efficient** — Designed for Render's 512 MB free tier. Cache capped at 300 entries with LRU eviction. Concurrent predictions use daemon threads, not asyncio tasks.
8. **Self-Improving** — Outcome logger tracks every prediction. IC engine retrains weekly. Factor weights evolve as the model sees more real-world outcomes.
9. **Real-Time Ready** — 15-minute prediction cache, async background computation, React Query polling for live data.
10. **Investor Transparency** — Every number in the UI is traceable to a specific calculation in the codebase. This document is kept current with every code change.
11. **Postgres-First Persistence** — All user data (watchlist, alerts, paper trades, picks, validation, alpha engine) is stored in PostgreSQL. Render's ephemeral disk is never trusted for user-facing state.

26. Changelog

Session 7 — 2026-06-22

Dashboard Tab Renamed "Gold & Silver" → "Commodities":

- More professional label, and leaves room to add more commodity instruments later (e.g. oil, platinum) without relabeling the tab again. `COMMODITY` internal key/type was already generic — only the two display strings (tab label, section heading) needed updating.

Crypto Fundamentals Check + Dead-End Tab Removal:

- Verified crypto's actual signal engine (`crypto_engine.py`) already correctly skips fundamentals entirely — technical + fear/greed (volatility proxy) + on-chain proxy (volume-based) only, explicitly documented as "no fundamentals." (An initial direct test of `PredictionEngine.predict()` for a CRYPTO market was misleading — that's dead code for crypto in production; `predictions.py`'s router actually dispatches CRYPTO to `predict_crypto()` instead.)
- The real issue was UI, not logic: the **Fundamentals tab** (screener.in data) rendered unconditionally on every stock detail page, but screener.in only covers Indian companies — for US, Crypto, and the tracking-only commodity ETFs it could only ever show "available for Indian (NSE) stocks only." Filtered it out of `HORIZON_TABS` for any market other than IN, since there's no scenario where it shows real data otherwise.

Gold & Silver Dashboard Tab:

- GLD/SLV/GOLDBEES/SILVERBEES were only reachable via search before this — no discoverable spot on the app's main landing page. Added a 4th market tab on the Dashboard (`frontend/src/app/dashboard/page.tsx`) next to India/USA/Crypto, showing live price + day change for all four as simple cards, same pattern as the existing Crypto tab's grid.
- No index bar (no natural index for 2 commodities — same simplification already made for Crypto) and no Quick Access/horizon-info sections (not meaningful for a fixed 4-symbol list) — just the price cards plus a note that these are tracking-only with no AI signal.

Gold/Silver Tracking Support — Fixed Fabricated Analysis:

- **Caught live:** after adding GLD/SLV/GOLDBEES/SILVERBEES to the universe, visiting their stock detail page produced a confident SELL signal whose bear case claimed *"Underperforming the Nifty 50 benchmark"* for a US gold ETF, plus invented valuation/earnings claims — despite `confidence_breakdown.data_completeness` being `0`. The fundamentals/sentiment/quality pipeline was fabricating plausible-sounding analysis from data that didn't exist.
- **Fixed in `PredictionEngine.predict()`:** added an early-return branch for `TRACKING_ONLY_SYMBOLS` (`stock_universe.py`) that computes the signal from real technical indicators only (RSI/MACD/EMA/ADX/volume — legitimate for any price series) and skips the entire fundamentals/sentiment/quality/bull-bear-case pipeline, returning a `tracking_only: true` flag and an honest explanatory note instead.
- **Frontend:** stock detail page now shows a clear "price-tracking instrument" banner and hides Bull/Bear Case, Factor Attribution, and Academic Quality Signals when `tracking_only` is set.
- **Bonus fix:** this surfaced two latent null-pointer crashes (`prediction.fundamental_score.score` / `sentiment_score.score` accessed without a null guard) that would have thrown for *any* symbol with missing fundamental data, not just the new commodity ETFs — fixed with optional chaining.
- Added `GLD`/`SLV` (US) and `GOLDBEES`/`SILVERBEES` (India) ETF tickers to both `backend/services/stock_universe.py` and `frontend/public/stock_universe.json`, so they're searchable and usable in Watchlist, Alerts, and the stock detail page (price + chart) via existing infra.
- **Deliberately scoped to tracking only** — no AI BUY/SELL signal, no entry in Daily Picks. Fundamental factors (P/E, ROE, Piotroski score) that drive equity scoring don't meaningfully apply to a commodity ETF; full signal support would need a separate technical+macro-only model, considered and explicitly deferred rather than built half-heartedly into the existing equity-scoring engine.
- Both lists are marked as manual additions outside the universe auto-generator's source (S&P/NASDAQ-100 + all NSE equities) — they'll need to be re-added if `scripts/generate_stock_universe.py` is ever re-run.

Full Indian Holiday Calendar + Muhurat Trading:

- **Gaps found:** the frontend's `marketHours.ts` was missing several *fixed-date* NSE holidays (Ambedkar Jayanti Apr 14, Maharashtra Day May 1, Christmas Dec 25, Good Friday) — it already had the Easter-algorithm machinery for US holidays but wasn't applying it to India. The lunar/regional holiday list (`NSE_EXTRA_HOLIDAYS`) was completely empty. The backend's market-hours check (duplicated in `paper_trading.py` and `screener_service.py`) had **zero** holiday awareness — not even the fixed ones.

• **Fixed:** added the missing fixed holidays (including Good Friday via the existing Easter computation) to `nseMarketHolidays()`; populated `NSE_EXTRA_HOLIDAYS` with the verified 2026 list (Holi, Ram Navami, Mahavir Jayanti, Bakri Eid, Moharram, Ganesh Chaturthi, Dussehra, Diwali, Guru Nanak Jayanti) sourced from NSE's official circular via Zerodha's mirror; wired it into the `nextEvent()` lookahead too (previously ignored).

• **Muhurat trading** — the special ~1hr evening session NSE/BSE run on Diwali Laxmi Pujan despite that date otherwise being a holiday — is now modeled via a `MUHURAT_SESSIONS` override list. 2026 falls on Sunday Nov 8; exact timing is a **placeholder** until NSE publishes the official window (~2 weeks before Diwali) — needs a follow-up update closer to that date.

• **Backend/frontend sync:** extracted a shared `backend/services/market_hours.py` mirroring the frontend logic exactly (same Easter computation, same 2026 extra-holiday list) and pointed both `paper_trading.py` and `screener_service.py` at it, removing two inline duplicates that were drifting out of sync with the frontend and with each other.

• **Operational note:** `NSE_EXTRA_HOLIDAYS` (both the TS and Python copies) needs a manual refresh every December for the following year, from NSE's official circular at nseindia.com/resources/exchange-communication-holidays.

Two Manuals Added:

• `StockSense360_Technical_Handbook.docx` — internal architecture/AI-engine/infra reference for engineers and the founder.

• `StockSense360_User_Guide.docx` — end-user/investor-facing walkthrough of every page, with 16 real screenshots captured from production (also doubles as a visual confirmation that this session's fixes shipped correctly).

• Both manuals later updated again in this session to reflect the market-hours gating, holiday calendar, Commodities tab, and tracking-only-instrument changes below (the manuals had drifted behind the markdown/PDF changelog).

US Market Support Added to Daily Picks:

• **Root question:** Daily Picks only ever covered NSE India — not a technical limitation, just never wired up. `PredictionEngine.predict()` already accepted a `market` parameter and worked correctly for US stocks (used on every US stock detail page); the daily-picks batch job (`daily_picks.py`) was the only piece hardcoded to `market="IN"`.

• **Backend:** generalized the bulk momentum screener (`_bulk_screen_nse` → `_bulk_screen(market, ...)`), the mcap-floor screener (NSE exchange code `NSI` vs US exchange codes `NMS`/`NYQ`/`NGM`/`ASE`/`PCX`), the regime-detection proxy ticker (`RELIANCE` → `AAPL` for US), the per-stock prediction call, and the disk/Postgres cache — all now take a `market` argument instead of assuming India. Added a 100-ticker US mega-cap fallback list (mirrors the existing Nifty 100 fallback role) for when the live screener call fails.

• **Postgres:** added a `market` column to `daily_picks_cache` (`ALTER TABLE ... ADD COLUMN IF NOT EXISTS`) so `save_picks_to_db`/`load_picks_from_db` keep independent history per market instead of one market overwriting the other's cache row.

• **API:** `/api/picks/daily`, `/api/picks/status`, and `/api/picks/generate` all accept a `market=IN|US` query param (default `IN`, validated against an allow-list). The `_generating`/`_last_error` module state in `daily_picks.py` changed from a single bool/string to a dict keyed by market, so an IN run and a US run can't trip each other's "already running" guard.

• **Cron:** GitHub Actions workflow (`daily_picks.yml`) now runs two scheduled jobs — India at 20:30 UTC (2 AM IST, unchanged) and US at 12:30 UTC (~8:30 AM ET, comfortably before the 9:30 AM ET open) — each calling `/api/picks/generate` with its own `market` param.

• **Frontend:** Daily Picks page (`/picks`) gets an IN/US toggle next to the existing horizon tabs. Switching markets changes the query key (separate cache per market), currency symbol (■/$), number locale (`en-IN`/`en-US`), display timezone for "Updated at," and the market passed through to the stock-detail link and the Paper Trade modal — all previously hardcoded to India.

• **Known limitation, not fixed in this pass:** the live picks-performance tracker (`/api/picks/performance`) and score-snapshot history key only on `(symbol, horizon)`, with no market column — if a US ticker and an NSE ticker ever share the same symbol, their historical performance rows could blend. Low real-world risk (ticker collisions across the two universes are rare) but worth a follow-up if it's ever seen in practice.

Market-Hours Gating for Paper Trading:

• **Root issue:** Buy/Sell could execute instantly at any time of day using a stale last-close quote when the market was closed — unrealistic (a real market can gap through that exact price by next open) and not a good look for a

tool positioning itself as a serious research platform.

• **Frontend:** `PaperTradeModal` now checks `getMarketStatus(market)` (the same utility already driving the navbar's market-status pills), shows a "{market} market is closed — opens at ..." banner, and disables the submit button while closed.

• **Backend:** mirrored the same check in `paper_trading.py`'s `/buy` and `/sell` endpoints (a local `_is_market_open()`, same pattern already used in `screener_service.py`) — a direct API call would otherwise bypass the frontend-only gate entirely.

Paper Trade Target/Stop-Loss Proximity Notifications:

• **New** `services/trade_notifier.py` — a background loop (every 15 min, registered in `main.py`) scans all OPEN paper trades with a `target_price` or `stop_loss` set, fetches the live quote, and emails the owner once price is within 2% of (or has crossed) either level. Each trigger is deduped via `target_notified_at`/`stop_notified_at` columns + a 6-hour cooldown, so a price hovering near the line doesn't spam the same email repeatedly.

• **Sends via Resend's HTTP API directly** (not through Supabase's SMTP) from `alerts@stocksense360.com` — a distinct sender from `invites@` so users can tell notification types apart. Requires a `RESEND_API_KEY` env var on the Railway backend service (reuses the same Resend account/API key set up for invite emails).

• `paper_portfolio.email` **column** stores the user's email captured from the frontend (`useAuth().user.email`) on every `/buy` and `/portfolio` call — no Supabase admin API call needed on the backend.

• **Browser popup notifications** — `OpenTradeRow` fires a `Notification()` once per (trade, kind) per session when price nears target/stop, gated on `Notification.permission === "granted"`. New "Enable Notifications" button added to the Paper Trading page header (browser permission prompts require a user gesture).

Daily Picks Generation Crash Fix:

• **Root cause found** — the 2 AM IST daily picks cron run crashed every single day on the short-term overbought-RSI quality gate in `daily_picks.py`: `" ".join(r.get("reasoning", []))` assumed `reasoning` was a list of plain strings, but it's actually a list of structured dicts (`{"indicator":..., "reason":...}`) built in `prediction_engine.py` for the factor-breakdown UI. The crash threw `TypeError: sequence item 0: expected str instance, dict found`, caught by the top-level crash handler, which silently saved an empty fallback payload (`{"short": [], "medium": [], "long": []}` + an `error` field) instead of real picks — so the Daily Picks page showed either a stale "Generating picks…" spinner (during catch-up retries) or "No BUY signals found today" (a misleading message; it wasn't that no signals existed, the run never got far enough to find any).

• **Fixed:** extract `item.get("reason", "")` from each dict before joining, with a defensive fallback for plain strings.

• **Verified live** via `/api/picks/status` — confirmed the crashed run's `error` field, then watched a fresh catch-up-triggered run complete with `last_error: null` after the fix deployed.

Daily Picks Target-Price / Upside Methodology (clarified, not a bug):

• Confirmed via `prediction_engine.py::_estimate_target()`: every **BUY** signal's target price has a hard-coded floor relative to current price — medium-horizon floors at `price * 1.05`, long-horizon at `price * 1.15`, short-horizon via an ATR-based move. This means **every BUY pick is guaranteed to show positive upside by design**, not because each stock's underlying analyst-target/trend math happened to be positive. If several BUY picks' natural projections fall below the floor, they'll cluster at exactly the same floored upside % (e.g. several stocks all showing "+5.0% upside" identically) — that's expected behavior, not a calculation bug, but worth knowing when interpreting the displayed upside numbers as "genuine" per-stock projections.

Validation Universe Bug — India Results Were Permanently Shadowed:

• **Root cause found** — `val_runs` had no `universe` column. `get_latest_results()` / `get_per_stock_results()` only filtered by `horizon`, picking whichever run had the highest `id` (i.e. most recent). Since the daily validation schedule always runs `nifty100` → `midcap` → `us` in that order, the **US run is always the most recent** for any given horizon — so the Validation page always displayed US results, and India (nifty100/midcap) results, though present in the database, were never shown.

• **Fixed:** added a `universe` column to `val_runs` (Postgres: idempotent `ALTER TABLE ADD COLUMN IF NOT EXISTS`, since the table already existed in production and `CREATE TABLE IF NOT EXISTS` alone is a no-op on existing tables; SQLite: `ALTER TABLE` wrapped to ignore "duplicate column" on repeat init). `get_latest_results()`/`get_per_stock_results()` now accept and filter by `universe` (default `nifty100`).

• **API:** `/api/validation/results`, `/results/stocks`, `/results/stock/{symbol}` all now accept a `universe` query param.

• **Frontend:** Validation page now has a Nifty 100 / Midcap / US selector above the horizon tabs. All "vs Nifty" benchmark text is now dynamic — shows "vs S&P 500" when viewing the US universe instead of incorrectly saying

Nifty for US data.

Railway Redeploy Scoping:

• **Root cause found** — Railway redeployed the backend service on every push to `main`, regardless of which files changed, including pure frontend and documentation commits. Each restart re-runs the startup "catch-up" check in `main.py`, which can kick off a brand-new ~10-15 minute full picks-generation run if it lands in a window before the day's legitimate 2 AM IST cron run has finished persisting — producing duplicate/wasted runs and a confusing "Generating picks…" spinner shown against an already-complete day's data.

• **Fixed:** added `railway.json` at repo root with `build.watchPatterns: ["backend/**"]`, so the backend service only redeploys when backend code actually changes. Frontend and docs-only pushes no longer restart it.

Invite Registration Fix:

• **Root cause found** — invite links (and password-reset links) authenticated the user for one Supabase session via magic-link code exchange, then dropped them straight onto `/accept-terms`. The user never set a password. On their next visit, `/login` only offers email + password sign-in with no Sign Up option (by design — invite-only app) — so an invited user with no password had no way back in.

• **New** `/auth/set-password` **page** — shown right after the invite/reset link authenticates the session; lets the user create a real password (min 6 chars, confirm match) via `supabase.auth.updateUser({ password })`, then continues to `/accept-terms`. Shows a clear "link expired" message if there's no active session.

• `/auth/callback/route.ts` **updated** — now redirects to `/auth/set-password?next=/accept-terms` instead of straight to `/accept-terms`. This covers both the invite flow and the forgot-password flow (previously forgot-password also had no way to actually set the new password after clicking the reset link).

• **Login page footer clarified** — explains invited users should look for an invite email with a link rather than expecting a Sign Up form.

• **Operational note:** Supabase Auth → URL Configuration must have `<site-url>/auth/callback` in the allowed Redirect URLs list for invite/reset links to work at all. If invites still fail after this fix, check that setting in the Supabase dashboard.

• **Production domain config fixed in Supabase** — Site URL updated from the default `*.vercel.app` to `https://stocksense360.com`; Redirect URLs now include `stocksense360.com`, `stocksense360.in`, `www.stocksense360.in`, and the vercel.app fallback, each with `/auth/callback`.

• **Second root cause found** — Supabase's dashboard **"Invite user"** button (and password-reset emails) don't support a custom redirect target; they always send the user to the bare **Site URL root** using the older **implicit/hash-based** flow: `https://stocksense360.com/#access_token=...&type=invite`. Hash fragments never reach the server, so the server-side `/auth/callback/route.ts` (which only handles `?code=...`) never even saw these — the user landed authenticated-but-unnoticed on the public homepage with no path forward.

• `InviteHashRedirect` **added to** `providers.tsx` — runs on every page load app-wide; checks `window.location.hash` for `access_token` + `type=invite`/`type=recovery`.

• **Third root cause found** — `@supabase/ssr`'s `createBrowserClient` (used in `lib/supabase.ts`) does **not** auto-detect/establish a session from hash-fragment tokens the way the classic `supabase-js` client does (no `detectSessionInUrl`). The first version of `InviteHashRedirect` only checked for the hash and redirected to `/auth/set-password` — it never actually called `setSession()`, so the page found no real session and would show "link expired" even on a valid, unused invite token.

• **Fixed:** `InviteHashRedirect` now parses `access_token` + `refresh_token` out of the hash with `URLSearchParams` and calls `supabase.auth.setSession({ access_token, refresh_token })` before navigating to `/auth/set-password?next=/accept-terms`.

• **Operational note — invite tokens are single-use:** clicking an invite/reset link consumes the token at Supabase's `/verify` endpoint regardless of whether the app does anything useful with the result. Any invite clicked before this fix shipped needs a **fresh invite resent** — the same link won't work twice.

Branded Invite Emails (Custom SMTP):

• **Custom SMTP via Resend** configured in Supabase → Authentication → Emails → SMTP Settings, sending from `invites@stocksense360.com` instead of Supabase's shared default sender — fixes deliverability and removes the "looks like spam" concern with invite emails.

• **Domain verification:** `stocksense360.com` added to Resend with DKIM (TXT), MX, and SPF (TXT) records added in GoDaddy DNS. Domain-level status can lag behind individual record checkmarks — re-trigger verification from the Resend domain detail page if status shows "Not Started" despite green record checkmarks.

• **Branded HTML invite template** (`supabase_invite_email_template.html` in repo root) pasted into Supabase's "Invite user" email template — dark themed, StockSense360 logo, "Team StockSense360" sender framing (no

individual name), clear one-time-link messaging.

• **Diagnosing SMTP failures:** Supabase's "Failed to invite user" toast doesn't show the real cause — query `auth_logs` in Logs → Explorer (`select cast(timestamp as datetime) as timestamp, event_message, metadata from auth_logs order by timestamp desc limit 10`) to see the actual GoTrue/SMTP error (e.g. domain not verified, bad credentials).

Per-User Terms Cookie Bug:

• **Root cause found** — the `ss_terms=v1.0` cookie set after accepting the Terms of Use disclaimer was **not scoped to a specific user** (`path=/` with a fixed name). Any user authenticating in a browser that had *previously* accepted terms as a *different* account would see the cookie, skip `/accept-terms` entirely, and land straight on `/dashboard` — never asked for name/mobile/country or shown the legal disclaimer.

• **Fixed in two places:** `accept-terms/page.tsx` (both the read-check and the write-after-accept) and `useAuthGuard.ts` (the app-wide route guard) now use a cookie scoped per user: `ss_terms_${user.id}=v1.0`. The guard fix matters more — it could previously let a user bypass the disclaimer-acceptance redirect entirely on a shared browser, not just skip the profile form.

Session 6 — 2026-06-20

User Feedback System:

• **Signal thumbs up/down** — users can rate each AI signal (BUY/HOLD/SELL) directly on the stock detail page. Widget renders below the Paper Trade button; shows current vote highlighted in bull/bear colour. Votes are upserted per `(user_id, symbol, market, horizon)` so toggling works cleanly.

• **Monthly NPS survey** — `NpsPopup` component appears globally (bottom-right) after a user has voted on at least one signal and then every 30 days thereafter. 0–10 score card + optional free-text comment. Colour-coded (green ≥9, yellow 7–8, red ≤6).

• **New backend router** (`backend/api/routers/feedback.py`) with 4 endpoints:

- `POST /api/feedback/signal` — upsert thumbs vote
- `GET /api/feedback/signal/{symbol}` — fetch user's existing vote
- `GET /api/feedback/signal/summary/{symbol}` — aggregate approval % across all users
- `POST /api/feedback/nps` — submit NPS score + comment
- `GET /api/feedback/nps/due` — returns `{due: bool}` based on 30-day cadence

• **DB schema additions** (already in `postgres_store.py` SCHEMA_SQL):

- `signal_feedback` table: per-user signal votes with UNIQUE constraint and ON CONFLICT upsert
- `nps_responses` table: per-user NPS scores with timestamps for 30-day cadence check

Look-ahead Bias Fix (Validation Engine):

• `_backtest_stock()` now recomputes indicators on a rolling window (`df.iloc[:i+1]`) at each signal date instead of on the full historical DataFrame. This eliminates future-price leakage into EMA-200, MACD, and OBV at time t.

• `MIN_WARMUP` raised 50 → 200 bars to ensure EMA-200 is valid before scoring begins.

• US symbols detected via `is_us` flag (`.NS` suffix no longer blindly appended).

• Validation hit rates will be modestly lower but accurately reflect real-time model performance.

Validation Coverage Expansion:

• Added `NSE_MIDCAP` universe (100 non-Nifty-100 NSE stocks) to `validation_engine.py`.

• Added `US_BASKET` universe (48 S&P 500 stocks spanning all GICS sectors).

• Railway cron (`_validation_schedule_loop` in `main.py`) now cycles all 3 universes (`nifty100`, `midcap`, `us`) back-to-back with 5-minute gaps between each.

• Deleted `daily_validation.yml` GitHub Action (was double-running alongside Railway cron).

• `/api/validation/run` endpoint accepts `universe` query param (`nifty100` | `midcap` | `us`).

Session 5 — 2026-06-20

User Identity & Persistence:

• **All features migrated to Supabase** `user_id` — watchlist, alerts, paper trading, and StockContextMenu all previously used a hardcoded `USER_ID = "default"` or a localStorage `session_id`. Every feature is now scoped to the authenticated Supabase user UUID (`useAuth().user.id`), making data persist correctly across all browsers and devices.

• **Paper trading backend rewritten** — all 5 API endpoints migrated from `session_id` (random localStorage UUID) to `user_id`. `BuyRequest`, `SellRequest`, `EditRequest` models updated; `_ensure_portfolio()` queries by `user_id`. Old trades with `user_id = NULL` are legacy-only.

• **Dashboard watchlist fixed** — was fetching `/api/watchlist/default`; now uses `userId` from `useAuth`.

Heatmap Expansion & Quality:

• **India: 18 → 25 sectors** — added Healthcare, Insurance, Chemicals, Cement, Metal & Mining, Defence, Realty, Telecom, Consumer Disc, Hotels & Travel, Food & Beverage, Media & Entmt, Textiles, Agro & Chemicals, Logistics, Paints, Infra, Capital Goods, Power, EV & New Energy

• **US: 13 → 29 sectors** — added Cybersecurity, Fintech, Biotech, Med Devices, Clean Energy, EV, Consumer Stap, E-commerce, Social Media, Streaming & Media, Gaming, Airlines, Cruise & Hotels, Restaurants, Retail, Telecom, Utilities, Realty, Materials, Crypto & Blockchain

• **MAX_STOCKS raised 10 → 15** — wider coverage per sector

• **Full symbol audit** — all 353 India heatmap symbols bulk-tested via Yahoo Finance. 37 bad symbols identified; confirmed replacements applied (ATGL, ASTERDM, BAYERCROP, CANFINHOME, DALBHARAT, GUJGASLTD, ICICIPRULI, KNRCON, LEMONTREE, MFSL, MTARTECH, NH, ORIENTCEM, TEJASNET, TIPSFILMS, VTL, VIJAYA, VINATIORGA, WAAREEENER, WELSPUNLIV, ZENSARTECH, ETERNAL). Symbols with no Yahoo Finance listing removed (SPICEJET, GREENKO, HEXAWARE, KEYSTONE, etc.)

• **ZOMATO → ETERNAL** — company rebranded to Eternal on NSE

• **NSE** `SECTOR_TO_NSE_INDEX` **expanded** — 13 new sector → NSE index mappings added for primary data sourcing via NSE APIs

UX Fixes:

• **Landing page auth confusion fixed** — page now detects login state via `useAuth`; CTA switches from "Sign In" to "Go to Dashboard" when logged in; bottom CTA shows user email

• **Loading status badge** added to Market Overview (dashboard) and Heatmap — three states: blue "Fetching…" (first load), yellow "Refreshing…" (background poll), green wifi "Updated HH:MM:SS" (live)

• **M%26M / URL-encoding fix** — NSE symbols with `&` (M&M, M&MFIN) were displaying as `M%26M` in the stock page header. Fixed with `decodeURIComponent` on the stock page params and `encodeURIComponent` when building `/stock/` URLs in heatmap, context menu, and picks pages

• **Paper Trade modal unblocked** — Buy button was disabled until `fetchPrediction` completed (10–30s). Prediction now runs in background for stop loss/target suggestions only; button is immediately available

Infrastructure:

• **Migrated backend from Render → Railway Hobby** ($5/month, always-on, no cold starts)

• `PICKS_CANDIDATES=751` set in Railway environment — expands the momentum screen candidate pool

Session 4 — 2026-06-19

Forensic Audit & Critical Fixes (9 issues resolved):

• **BUY threshold lowered 70 → 60** — the 70-point threshold was structurally unreachable for most NSE stocks on neutral/bearish market days. New thresholds: BUY ≥ 60, HOLD 45–59, SELL < 45. Score bands updated to match exactly (no more "Good Watchlist Stock" label on stocks treated as HOLD).

• **Fundamental score per-category budgets** — replaced unbounded additive accumulation (theoretical max ~215) with six capped buckets: valuation ±15, profitability ±15, growth ±15, balance sheet ±10, governance ±10, banking ±10. Scores now discriminate meaningfully across the full 0–100 range.

• **Growth double-counting fixed** — revenue and earnings growth were previously counted 3–5× simultaneously (TTM + 3Y CAGR + 5Y CAGR + trend, all additive). Now each counted once via the longest available window; quarterly trend is a capped supplement.

• **Quality gate fixed** — OCF check used Python `or` which treated zero cash flow as falsy; now uses explicit `is None`. Rejection logic split into independent OR conditions (was AND — too strict).

• **Sentiment denominator corrected** — neutral articles no longer dilute bullish signal. Denominator is now `bullish + bearish` only.

- **Race condition on `_generating` flag** — concurrent POST `/picks/generate` requests could both pass the guard simultaneously. Fixed with `threading.Lock`.
- **Optimizer fallback max_weight enforcement** — the fallback could allocate 100% to one stock when alphas were skewed. Fixed with iterative clipping loop.
- **Long-horizon IC training** — long horizon was training on 20D returns (1 month) despite a 3-6 month stated horizon. Now uses `return_60d`. Outcome logger waits 90 calendar days before resolving long-horizon predictions.
- **Partial returns never logged** — outcome logger previously logged partial forward returns when the window hadn't elapsed, corrupting IC training data. Now returns `None` until the full window is complete.
- **Validation thresholds synced** — validation BUY threshold (was 65) now matches live system (60) for all horizons.
- `picks_generated_today()` **logic fixed** — a prior run that produced 0 BUY signals (before threshold fix) saved an empty payload with today's date, causing the startup catch-up to skip regeneration. Now requires at least one actual pick to count as "done today".
- **NSE Daily Picks expanded** — universe expanded from Nifty 100 (96 stocks) to all NSE-listed stocks screened by market cap ≥ ■100 Cr (~500-600 stocks). Two-phase pipeline: Phase-0 bulk momentum screen → Phase-1 deep prediction on top 50 candidates only (memory-safe batching of 300).
- **Documentation fully updated** — README and STOCKSENSE_DOCUMENTATION.md updated to reflect all threshold, formula, and architecture changes.

Live test results (2026-06-19):

- `/health`: ■ ok
- `/api/predictions/TCS?market=IN&horizon=medium`: ■ Score 78, BUY, Strong Buy Candidate
- `/api/screener/top-movers?market=IN`: ■ 10 gainers, 10 losers
- `/api/validation/results?horizon=medium`: ■ Hit rate 56.6%, avg return 3.75%
- `/api/picks/status`: ■ Generating (startup catch-up triggered correctly)

Session 3 — 2026-06-18

New Features:

- **Paper Trading module** — full simulated trading with open/close positions, stop-loss/target tracking, unrealised and realised P&L, Postgres persistence
- **Price Alerts system** — Postgres-backed price level alerts with live trigger detection
- **Daily Picks Trust Layer** — real backtest results, confidence calibration table per score band, live P&L tracker for past picks
- **Pick Card UI overhaul** — rank badges (#1–#5), sector tags, top 3 signals visible inline, compact regime bar

Fixes:

- Watchlist migrated from ephemeral JSON file to Postgres — no longer disappears on Render restart
- screener.in login now fires at startup (not lazily) — data available from first request
- screener.in login enhanced: tries both `username`/`email` field names, logs every step
- Daily picks cron now waits for Render to be healthy before triggering — prevents silent cold-start failures
- Paper trading open positions: stop loss/target hint row now always visible (shows "not set — click ✎ to add one" when unset, for consistent layout)
- Open positions now always show SL/target hint row consistently regardless of signal type

Performance Improvements:

- `get_fundamentals`: replaced 3×sleep(3) blocking retry with `asyncio.wait_for(timeout=8s)`
- OHLCV data: added 5-minute in-process cache (was completely uncached)
- Quote enrichment: removed redundant second yfinance `fast_info` call
- Heatmap cache TTL: 3 min → 5 min
- Dashboard refetch interval: 60s → 120s, staleTime aligned
- Heatmap frontend refetch: 3 min → 5 min, staleTime aligned
- Stock detail quote refetch: 30s → 60s
- `refetchOnWindowFocus: false` added across all major queries

Session 2 — (prior)

- Walk-forward validation engine with Postgres storage
- Learning Alpha Engine (IC weights, regime clustering, meta-model, outcome logger)
- Screener.in authenticated scraping for Indian fundamental data
- Portfolio page with BUY/HOLD/SELL signals per holding
- History tab with horizon selector on portfolio page
- Backtest async fix (non-blocking event loop)
- Prediction cache size cap (300 entries, LRU eviction)
- Hammer & Morning Star candlestick pattern bug fixes

Session 1 — (initial)

- Core prediction engine (technical, fundamental, sentiment, quality, macro)
- Stock detail page with full factor breakdown
- Dashboard with top movers (US/IN/Crypto)
- Heatmap page (sector-wise colour-coded)
- Screener with filters
- Watchlist with live prices
- Daily Picks engine (9-phase Learning Alpha pipeline)
- GitHub Actions automation (daily picks, weekly validation, keep-alive)
- Render + Vercel deployment

This document is a living record of StockSense. It is updated with every significant change to the product.